

MANIWAKI, QC, Canada

WMO: 717210

Lat: 46.2736N

Lon: 75.9878W

Elev: 200

StdP: 98.95

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-25.0	-32.5	0.2	-27.5	-29.4	0.3	-24.1	7.7	-4.9	6.9	-5.7	1.2	330	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.9	20.3	28.1	19.5	26.5	18.8	22.5	26.8	21.5	25.4	20.5	24.2	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.3	24.1	20.2	15.2	23.3	19.2	14.3	22.4	67.4	26.8	63.4	25.4	60.0	24.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
6.8	6.0	5.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-32.3	32.6	3.0	1.6	-34.5	33.8	-36.2	34.8	-37.9	35.7	-40.1	36.9		
			-32.4	24.4	2.9	1.4	-34.5	25.4	-36.2	26.2	-37.9	27.0	-40.0	28.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-11.8	-9.8	-3.7	4.2	12.0	17.0	19.2	18.1	13.9	7.1	0.0	-7.3	(6)	
(7)		DBStd	12.04	7.69	6.67	6.42	4.89	4.55	3.84	2.94	3.17	4.28	4.55	5.32	6.73	(7)	
(8)		HDD10.0	2835	677	555	424	183	30	1	0	0	12	115	301	537	(8)	
(9)		HDD18.3	5007	936	788	682	424	203	69	24	44	144	348	550	795	(9)	
(10)		CDD10.0	1007	0	0	1	10	92	212	287	252	128	25	1	0	(10)	
(11)		CDD18.3	136	0	0	0	0	7	29	52	37	11	1	0	0	(11)	
(12)		CDH23.3	1332	0	0	1	8	109	323	482	309	96	4	0	0	(12)	
(13)		CDH26.7	331	0	0	0	1	26	91	126	66	22	0	0	0	(13)	
(14)	Wind	WSAvg	2.2	2.2	2.3	2.5	2.6	2.4	2.1	2.0	1.9	2.0	2.3	2.3	2.1	(14)	
(15)	Precipitation	PrecAvg	904	65	49	62	62	68	90	91	94	91	79	77	72	(15)	
(16)		PrecMax	1099	129	117	143	106	118	147	174	191	181	133	181	127	(16)	
(17)		PrecMin	734	25	18	16	20	30	41	37	21	29	38	16	26	(17)	
(18)		PrecStd	104	26	23	34	24	27	31	42	36	40	23	33	28	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.2	15.9	24.1	29.8	31.9	32.3	31.1	29.4	23.1	15.2	8.6	(19)	
(20)		MCWB	7.1	4.2	9.8	13.9	17.9	20.9	22.1	21.1	20.7	16.4	12.5	7.1	7.1	(20)	
(21)		2%	DB	3.8	4.1	10.3	19.1	26.4	29.2	29.7	28.5	25.9	19.5	12.1	5.1	(21)	
(22)		MCWB	3.1	2.1	5.7	10.1	15.8	19.6	20.7	20.1	19.9	14.6	10.1	4.1	4.1	(22)	
(23)		5%	DB	1.4	2.1	7.4	15.7	23.5	27.1	27.8	26.6	23.6	16.9	9.8	3.1	(23)	
(24)		MCWB	0.7	0.6	3.6	8.9	14.2	18.7	19.6	19.3	18.0	13.0	8.0	2.1	2.1	(24)	
(25)		10%	DB	-0.6	0.2	5.0	12.6	20.7	24.8	26.1	24.9	21.4	14.4	7.5	1.3	(25)	
(26)		MCWB	-1.4	-0.8	2.2	6.8	13.4	17.4	19.0	18.5	17.4	11.6	5.7	0.5	0.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	4.9	11.4	15.8	20.7	23.6	24.2	23.2	22.4	18.4	13.3	7.6	(27)	
(28)		MCDWB	7.4	5.8	14.8	20.6	26.6	28.5	29.1	27.2	26.3	20.3	14.3	8.1	8.1	(28)	
(29)		2%	WB	3.2	2.5	6.6	12.6	18.2	21.7	22.7	21.9	20.8	15.6	10.5	4.4	(29)	
(30)		MCDWB	3.6	3.6	8.9	17.0	23.2	26.6	26.9	25.5	23.9	18.9	11.8	5.1	5.1	(30)	
(31)		5%	WB	0.7	0.8	4.2	9.5	16.4	20.4	21.5	20.8	19.4	13.7	8.2	2.3	(31)	
(32)		MCDWB	1.3	1.9	6.8	14.1	20.7	24.5	25.3	24.3	22.5	16.2	9.4	3.0	3.0	(32)	
(33)		10%	WB	-1.4	-0.8	2.4	7.5	14.6	19.0	20.4	19.7	17.9	11.8	5.9	0.7	(33)	
(34)		MCDWB	-0.7	0.4	4.7	11.7	18.9	23.0	24.0	23.2	20.7	14.2	7.3	1.2	1.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.0	11.0	10.8	11.0	12.3	11.8	11.6	11.5	11.3	9.2	7.7	8.2	8.2	(35)	
(36)		5% DB	MCDBR	8.8	10.5	12.4	17.3	16.8	15.1	14.1	13.7	13.9	13.0	10.7	7.5	7.5	(36)
(37)		MCWBR	8.6	9.1	8.3	9.8	7.7	6.8	6.6	6.5	7.8	8.3	8.9	7.1	7.1	(37)	
(38)		5% WB	MCDBR	8.7	9.2	11.3	14.4	13.0	12.2	11.5	11.1	11.9	11.4	10.0	7.4	7.4	(38)
(39)		MCWBR	8.6	8.4	8.3	9.5	7.4	6.7	6.4	6.2	7.4	8.2	9.1	7.2	7.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.262	0.281	0.316	0.341	0.371	0.396	0.396	0.383	0.365	0.334	0.302	0.269	0.269	(40)	
(41)		taud	2.329	2.253	2.282	2.403	2.377	2.329	2.333	2.383	2.414	2.502	2.501	2.366	2.366	(41)	
(42)		Ebn at Noon	844	894	910	917	895	871	865	864	848	830	796	793	793	(42)	
(43)		Edh at Noon	74	98	112	110	117	123	121	111	99	78	64	65	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.33	2.29	3.69	4.60	5.12	5.63	5.56	4.88	3.78	2.21	1.29	0.97	0.97	(44)	
(45)		RadStd	0.20	0.30	0.35	0.39	0.47	0.40	0.49	0.26	0.35	0.23	0.16	0.10	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page